United States Patent Application Publication 20250270124
Kind Code A1
Publication Date August 28, 2025
Inventor(s) Clark; Steve L

SYSTEM AND PROCESS FOR TURNING WASTE STREAMS INTO ELECTRICAL POWER

Abstract

A process for treating waste materials and generating electrical power from simultaneously comprising reacting the waste materials during a reaction with fuel, oxygen and water, and then oxidizing the gaseous reaction product of those materials along with fuel, oxygen and water. In one embodiment the process further comprises the steps of electrolyzing the water exiting the process to produce hydrogen and oxygen, purifying both the hydrogen and oxygen streams, and then feeding the purified hydrogen and oxygen to hydrogen fuel cells to generate power.

Inventors: **Clark; Steve L (Baytown, TX)**

Applicant: **Clark; Steve L** (Baytown, TX)

Family ID: **1000008280509**

Appl. No.: **18/883023**

Filed: **September 12, 2024**

Related U.S. Application Data

parent US division 18589830 20240228 PENDING child US 18883023

Publication Classification

Int. Cl.: **C02F1/58** (20230101); **C02F1/38** (20230101); **C02F1/72** (20230101); **C02F11/06** (20060101); **C02F101/14** (20060101); **C02F101/36** (20060101); **C25B1/04** (20210101)

U.S. Cl.:

CPC **C02F1/583** (20130101); **C02F1/38** (20130101); **C02F1/727** (20130101); **C02F11/06** (20130101); **C25B1/04** (20130101); C02F2101/14 (20130101); C02F2101/36

(20130101); C02F2209/02 (20130101); C02F2209/03 (20130101); C02F2301/046 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application claims priority to U.S. Utility application Ser. No. 18/589,830 (“the '830 Application”) filed Feb. 28, 2024. The '830 Application is hereby incorporated by reference in its entirety for all purposes, including but not limited to, all portions describing the waste treatment system disclosed, those portions describing waste treatment and power generation in general as background and for use with specific embodiments of the present invention, and those portions describing other aspects of the waste treatment and power generation system that may relate to the present invention.
STATEMENTS REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT
[0002] Not Applicable.
REFERENCE TO A MICROFICHE APPENDIX
[0003] Not Applicable.
BACKGROUND OF THE INVENTION
1. Field of the Invention
[0004] The present invention relates to a per and polyfluoroalkyl substance mitigation. More particularly, the present invention relates to a process for breaking down and destroying PFAS in waste streams. Even more particularly, the present invention relates to breaking down and destroying PFAS in a combustion process designed to have little or no harmful emissions.
2. Description of the Related Art
[0005] Per- and polyfluoroalkyl substances (PFAS) are a group of manufactured chemicals that have been used in industry and consumer products since the 1940s because of their useful properties. There are thousands of different PFAS, some of which have been more widely used and studied than others. Perfluorooctanoic Acid (PFOA) and Perfluorooctane Sulfonate (PFOS), for example, are two of the most widely used and studied chemicals in the PFAS group. PFOA and PFOS have been replaced in the United States with other PFAS in recent years.
[0006] PFAS can be present in our water, soil, air, and food as well as in materials found in our homes or workplaces, including in public drinking water systems and private drinking water wells, at landfills, disposal sites, and hazardous waste sites fire extinguishing foams, manufacturing or chemical production facilities that produce or use PFAS such as chrome plating, electronics, and certain textile and paper manufacturers, fish caught from water contaminated by PFAS and dairy products from livestock exposed to PFAS, food packaging, household products such as stain and water-repellent used on carpets, upholstery, clothing, and other fabrics; cleaning products; non-stick cookware; paints, varnishes, and sealants, certain shampoo, dental floss, and cosmetics, and fertilizer from wastewater treatment plants that is used on agricultural lands.
[0007] Current scientific research suggests that exposure to certain PFAS may lead to adverse health outcomes. Research is underway to better understand the health effects associated with low levels of exposure to PFAS over long periods of time, especially in children. Current peer-reviewed scientific studies have shown that exposure to certain levels of PFAS may lead to reproductive effects such as decreased fertility or increased high blood pressure in pregnant women, developmental effects or delays in children, including low birth weight, accelerated puberty, bone variations, or behavioral changes, increased risk of some cancers, including prostate, kidney, and testicular cancers, reduced ability of the body's immune system to fight infections, including reduced vaccine response, interference with the body's natural hormones, and increased cholesterol

levels and/or risk of obesity.
[0008] In 2021, the United States Environmental Protection Agency (EPA) established the EPA Council on PFAS and charged it with developing an ambitious plan of action to further the science and research, to restrict these dangerous chemicals from getting into the environment, and to immediately move to remediate the problem in communities across the country. This group released the “PFAS Strategic Roadmap: EPA's Commitments to Action 2021-2024” which is hereby incorporated by reference for all purposes including defining the problem, characteristics and sources of PFAS. In the EPA's roadmap, it identifies three technologies that are commercially available to either destroy or dispose of PFAS and PFAS-containing materials but raises concerns about significant uncertainties and information gaps that exist concerning the technologies’ ability to destroy or dispose of PFAS while minimizing the migration of PFAS to the environment.
[0009] Inventor Steve Clark's U.S. Pat. No. 5,906,806 entitled “Reduced Emissions Combustion Process with Resource Conservation and Recovery Options ‘ZEROS’ Zero-Emission Energy Recycling Oxidation System” based upon a Oct. 16, 1996, filing introduced a technology for combusting hydrocarbon streams to recover energy and maximize carbon dioxide production and recovery. As the thought leader of combustion and environmental cleanup, Clark continued his research and development to improve the ZEROS technology and to find new uses. Examples of Clark's continued progress in this technology area are found in his other issued patents which include U.S. Pat. No. 6,137,026 entitled “ZEROS Bio-dynamic a Zero Emission Non-Thermal Process for Cleaning Hydrocarbons from Soils”; U.S. Pat. No. 6,024,029 entitled “Reduced Emission Combustion System”; U.S. Pat. No. 6,119,606 entitled “Reduced Emission Combustion Process”; U.S. Pat. No. 6,688,318 entitled “Process for Cleaning Hydrocarbons from Soils”; U.S. Pat. No. 7,338,563 entitled “Process for Cleaning Hydrocarbons from Soil: U.S. Pat. No. 7,833,296 entitled “Reduced-Emission Gasification and Oxidation of Hydrocarbon Materials for Power Generation”; U.S. Pat. No. 8,038,744 entitled “Reduced-emission Gasification and Oxidation of Hydrocarbon Materials for Hydrogen and Oxygen Extraction; and U.S. Pat. No. 8,038,746 entitled “ ” Reduced-emission Gasification and Oxidation of Hydrocarbon Materials for Liquid Fuel Production.” The patents mentioned in this paragraph are hereby incorporated by reference in their entirety for all purposes, including but not limited to those portions describing the method, structure and technical details of the process as background and for use as portions of specific embodiments of the present invention.
[0010] As the EPA raised issues with the known process for destroying or disposing of PFAS, it would be desirable to have a process that reliably destroys PFAS from waste materials. It would be even more desirable to adapt a process which has little or no emissions to treat PFAS containing sludges and destroy the PFAS contained therein.

BRIEF SUMMARY OF THE INVENTION

[0011] The present invention is a process for treating PFAS containing waste materials comprising vaporizing the PFAS containing waste materials during a reaction with fuel, oxygen and water, and then oxidizing the gaseous reaction product of those materials along with fuel, oxygen and water to break the fluorine bonds and oxidize the remaining components to carbon dioxide and water. In one embodiment the process for treating PFAS containing waste materials comprises introducing a PFAS containing waste stream, a first oxygen stream, a first fuel stream and a first water stream into a primary combustion chamber operated under vacuum conditions; vaporizing the waste, oxygen, fuel and water streams in the primary combustion chamber to produce gaseous product stream comprising hydrogen, carbon dioxide, PFAS, water vapor and other hydrocarbons and a solids stream; introducing the gaseous product stream into a cyclone separator to separate gaseous components streams from retained particulate components, thereby creating a clean gaseous product stream; introducing the clean gaseous product stream into a secondary combustion chamber along with a second fuel stream, a second oxygen stream, and a second water stream, and thermally oxidizing the introduced components to produce a reaction product stream comprising carbon dioxide and water; recycling a portion of the reaction product stream to the secondary combustion chamber for additional oxidation of any uncombusted reactants; introducing the remaining portion of the reaction product stream into an acid wash scrubber to remove residual fluorine, thereby creating a scrubbed reaction product stream; and separating the scrubbed reaction product stream into a brine stream comprising water and a scrubbed gaseous stream comprising carbon dioxide.

[0012] In another embodiment of the process, the process further includes the steps of removing a portion of the water stream from the brine stream creating a regenerated water stream; introducing the regenerated water stream into an electrolysis unit thereby electrolyzing the regenerated water stream to produce a hydrogen and oxygen containing stream; separating the hydrogen from the oxygen in the hydrogen and oxygen containing stream thereby creating a purified hydrogen stream and a purified oxygen stream; and introducing the purified hydrogen stream and the purified oxygen stream into a hydrogen fuel cell to generate electric power.

[0013] Additional advantages of the invention are set forth in part in the description which follows, and in part will be obvious from the description, or may be learned by practice of the invention. The advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0014] A better understanding of the present invention can be obtained when the following detailed description of the disclosed embodiments is considered in conjunction with the following drawings in which:

[0015] FIG. **1**A/B is a process flow diagram of process for breaking down and destroying PFAS; and

[0016] FIG. **2** is a process flow diagram of a process for breaking down and destroying PFAS which also generates power through hydrogen fuel cells.

DETAILED DESCRIPTION OF THE INVENTION

[0017] The present invention is a process for treating PFAS containing waste materials comprising vaporizing the PFAS containing waste materials during a reaction with fuel, oxygen and water, and then oxidizing the gaseous reaction product of those materials along with fuel, oxygen and water to break the fluorine bonds and oxidize the remaining components to carbon dioxide and water.

[0018] FIGS. **1**A and **1**B illustrate a preferred implementation of the process and system comprising a primary gasification chamber **10** and a secondary oxidation chamber **20** used to gasify and oxidize, respectively, a PFAS containing waste stream **23** and a hydrocarbon fuel stream **21** to carbon dioxide, water, PFAS free solids and energy. As shown in FIG. **1**A, the process of a preferred embodiment of the invention begins by introducing feedstock stream **21**, PFAS containing waste stream **23**, oxygen stream **22**, and water stream **24** into primary combustion chamber **10**. Fuel stream **21** can be a variety of hydrocarbon feedstocks, including natural gas, hydrocarbon liquids such as butane, and other hydrocarbon-containing compounds. PFAS containing waste stream **23** can be a variety of sources such as sludges and biosolids from chemical plants, refineries and wastewater treatment facilities. The PFAS containing waste stream **23** can be delivered via vacuum trucks and fed directly to the process or stored in intermediate tanks. In general, the PFAS containing waste stream comprises between about 80% and 99% water. Preferably, waste stream **23** is fed into primary chamber **10** with an auger **25** or other similar type mechanism that allows the stream to be introduced without breaking the vacuum environment.

Water containing PFAS can also be injected into the primary chamber **10** (as well as the secondary or cyclone) where the water will be turned to steam and ultimately heated to split the fluorine molecule off. Within primary combustion chamber **10**, the hydrocarbon feedstock **21** and hydrocarbons in the PFAS containing waste stream **23** are converted to carbon dioxide, methane, carbon monoxide and hydrogen via the following three principal **55** chemical reactions, listed in order by the preferential affinity of carbon to oxygen in view of all other possible combustion reactions:

TABLE-US-00001 Primary Combustion Chamber C + O .fwdarw. CO2 Exothermic C + 2H2 .fwdarw. CH4 Exothermic C + H2O .fwdarw. CO + H2 Endothermic

[0019] The preferred internal operating conditions of the primary combustion chamber **10** comprise a 5% to 10% oxygen rich (i.e., excess oxygen) atmosphere with a preferred temperature of approximately 950 to 1050° F. and an internal pressure of about 8 to 11 psia (i.e., below atmospheric pressure). Preferably primary combustion chamber **10** is a rotary kiln having liquid (preferably water) seals (not shown) on each end to keep the generally liquid PFAS containing waste stream **23** inside the rotating kiln. Water is pumped to the seals to allow water to seep into the primary combustion chamber instead of air. Preferably, primary combustion chamber **10** is a multi-pass system to adequately evaporate the large water volume from the remaining solids. As an additional safety feature to enhance the safety associated with the process, primary combustion chamber **10** is connected to emergency vacuum chamber **124**. Primary combustion chamber **10** also has an solids separation section **60** (preferably slag funnels) for removing a portion of solid components including slag, glass and/or ash that results from the gasification process. In general, the solids **61** separated in the solids separation section **60** is about 0.01% of the total feed mass into primary combustion chamber **10**. Gasification product **28** is then introduced into a separation cyclone **62** to remove additional ash and solids. In general, about 99.99% of the mass volume introduced into the primary combustion chamber will exit in stream **28** to separation cyclone **62**. Preferably stream **28** represents no less than 98% of the combined mass of the PFAS containing waste stream **23**, the fuel stream **21**, the oxygen stream **22**, and the water stream **24**. Given the high gas volumes containing some solids generated in primary combustion chamber **10** from PFAS containing waste stream **23**, separation cyclone **62** preferably has a different design from prior art systems. Preferably, cyclone **62** comprises a enhanced thickness zone having as much as 10 times the steel thickness as prior art to prevent erosion caused by the large amount of solids in a sandblasting effect. Otherwise, separation cyclone **62** is of a variety commonly known to those skilled in the art of combustion process.

[0020] After the solids are removed, gasification product stream **28** is then sent to the secondary combustion chamber **20** via line **31**. Preferably, as shown in FIG. **1**A, secondary combustion chamber **20** is a vertical combustion chamber such as is known by those of ordinary skill in the art. While the secondary combustion chamber of the prior art was a large energy source, the current process utilizes the secondary combustion system **20** to thermally oxidize PFAS in the gasification product stream **28** and break the fluorine off the hydrocarbon molecule. Gasification product stream **28** containing PFAS and other remaining volatiles are reacted with an additional feedstock stream **30**, a second substantially pure oxygen stream **32**, and/or a second water stream **34** in secondary combustion chamber **20**. Preferably, gasification product stream **28** is preheated in exchanger **63**. Feedstock stream **30** can be a variety of hydrocarbon feedstocks, including natural gas, and other high BTU quality wastes. Preferably, secondary combustion chamber **20** operates below atmospheric pressure, most preferably between 10 psia and 12 psia. The preferred internal operating conditions of secondary combustion chamber **20** comprise a 5% oxygen rich atmosphere with a temperature of approximately 2000 to 2,400° F. This condition causes stoichiometric oxidation resulting in a synthetic air environment of carbon dioxide and water without damaging the metal chamber. Preferably, the secondary combustion chamber has about an 8 to 12 second hold time for the combined reacting streams, most preferably about 10 seconds hold time. The formation of carbon dioxide and water (i.e., steam) in the secondary combustion chamber **20** is an auto-thermal driven process that can be summarized by the following three principal chemical reactions, listed in order by the preferential affinity of carbon to oxygen in view of all other possible combustion reactions of the gases produced in the primary combustion chamber **10**: TABLE-US-00002 Secondary Combustion Chamber 2CO + O2 .fwdarw. CO2 Exothermic Reaction 2H2 + O2 .fwdarw. 2H2O Exothermic Reaction CH4 + 2O2 .fwdarw. CO2 + 2H2O Exothermic Reaction

[0021] Reaction product stream **38**, consisting primarily of carbon dioxide and water, exits from the top of secondary combustion chamber **20**. Solids, ash, and other particulate matter are removed from a bottom cone section **64** of secondary combustion chamber **20**. Secondary combustion chamber **20** is included in the process to produce high combustion efficiency and break the PFAS fluorine bound from the hydrocarbon.

[0022] In preferred embodiments inductor **35** such a large fan system is utilized to induce a circular draft through the secondary combustion chamber **20** with the discharge stream **37** leaving inductor **35** being returned to primary combustion chamber **10** for reprocessing. The inductor **35** allows for gaseous circulation through the system even when the system is not being fired.

[0023] An optional feature of the overall process is recovering energy, in the form of heat, from reaction product stream **38** leaving the secondary combustion chamber **20**. Preferably, an energy recovery boiler **14** is used to recover the heat energy from reaction product stream **38**. As those skilled in the art will recognize, energy recovery boiler **14** is used to generate steam by transferring the heat energy from reaction product **38** to a boiler feedwater stream **134** from boiler feedwater pre-heater **138**. A portion of stream **38** can be used in parallel with energy recovery boiler **14** to heat other process streams through heat integration (i.e., cross exchanges of energy). Alternatively, other types of heat exchangers (not shown) can be used to recover the heat energy from reaction product stream **38** in place of energy recovery boiler **14**. Removal of the heat energy from stream **38** in recovery boiler **14** results in a cooler stream temperature of approximately 1,200° F. Preferably, stream **38** is cooled to about 450° F.

[0024] Cooled reaction product stream **40** is then introduced into a bag house **66** for removal of particulate matter from cooled reaction product stream **40**. Bag house **66** is of a design commonly known and used by those skilled in the art. Preferably, an activated carbon injector **68** can be utilized along with bag house **66** to assist in removal of particulate matter. Upon exiting bag house **66**, product stream **41** is introduced into combustion gas manifold **70**. Fan **72** can be used to increase the pressure of product stream **41** prior to introduction of product stream **41** into gas manifold **70**.

[0025] In gas manifold **70**, product stream **41** is split into three streams. Stream **42**, containing the bulk of the flue gas, is routed to gas polishing **16** and purification/recovery 18 units. The remaining two streams **26** and **36** can be recirculated to the primary **10** and secondary **20** combustion chambers, respectively, to maintain the gasification/oxidation environment and increase the combustion efficiency. Stream **26** is recirculated to primary combustion chamber **10** through activated carbon filter **78**. Likewise, stream **36** is recirculated to secondary combustion chamber **20** through activated carbon filter **78**. The amount of recirculating combustion gas introduced into primary combustion chamber **10** is controlled by control valve **74** or other means of regulating flow volume. Similarly, the amount of recirculating combustion gas introduced into secondary combustion chamber **20** is controlled by control valve **76** or other means of regulating flow volume. Preferably, the temperature of the recirculated flue gas is reduced to approximately 175° F. just prior to the gas being reintroduced into the primary **10** and secondary **20** combustion chambers. To control the system and process, the primary **10** and secondary **20** combustion chambers are monitored for their specific oxygen saturation while flow controllers **74**, **76** are used to regulate the recirculation and thereby adjust oxygen levels in order to achieve maximum efficiency. This rigorous control, particularly of oxygen levels, virtually eliminates the production of dioxin within the system. The activated carbon filter **78** within recirculated flue gas streams **26**, **36** is a preferred feature of one or more preferred embodiments of the invention. When an additional carbon source is available and the recirculated flue gases in streams **26**, **36** are at or above 450° F., carbon dioxide present in the flue gas is converted to carbon monoxide. The carbon monoxide is generated through the Boudouard reaction (C+CO2.fwdarw.2CO) from the additional carbon available in the activated carbon filter **78** and the carbon dioxide present in recirculated flue gas streams **26**, **36**. The additional carbon monoxide generated increases the overall energy production and efficiency of system/process. Because waste residual heat is used to carry out the endothermic Boudouard reaction, no negative loss in heat gain is experienced in the primary **10** or secondary **20** combustion chambers. The amount of carbon consumed as a filter medium in activated carbon filter **78** is determined by the mass flow rate of recirculated flue gas which is further determined by the total gas flow rate of the system/process. Thus, the carbon within activated carbon filter **78** is a continuous feed system, similar to the reactant in a scrubbing system. While activated carbon filter **78** is shown in FIG. **1**A as being a single unit, separate filter units may be employed for each of the streams **26**, **36**. Alternatively, the activated carbon filter unit **78** may be employed on only one of the streams **26**, **36**.

[0026] Acid gases will not buildup if the temperature is maintained above the acid gas dew point. Thus, the recirculated flue gas temperature is preferably maintained between 450 to 485° F. to eliminate the problem associated with the build up of acid gases. The water in the recirculated gas streams **26**, **36** has the effect of moderating the internal temperature as well as providing a mechanism for the removal of sulfurs or metals from the system. The water in the recirculated gas streams **26**, **36** also provides a mechanism for the removal of acid buildup, such as hydrochloric acid buildup, formed during the oxidation of halogenated feedstocks.

[0027] As previously mentioned, the bulk portion of reaction product stream **41** exits combustion gas manifold **70** as stream **42**. Stream **42** comprises carbon dioxide, water, and various other impurities and unreacted components from the combustion process. As illustrated in FIG. **1**B, stream **42** is next introduced into acid scrubber system **86** to remove any remaining acidic constituents in the gas stream. Acid scrubber system **86** comprises an adiabatic quench **88** and pack bed absorber **90**. Acid scrubber system **86** is of a design commonly known to those skilled in the art of purifying gas streams. Pack bed absorber **90** employs an alkaline stream **92** in a countercurrent flow **30** arrangement to neutralize any acidic components within stream **42**. Optionally, acid scrubber system **86** may comprise a series of pack bed absorbers **90** to increase contact efficiency. The brine stream **94**, which results from a contact of the alkaline stream **92** with the acid gas components, can then be filtered in filtration system **96**. Stream **94** is concentrated in distillation brine concentrator **98** to produce, for example, a marketable 42% brine stream for use in downhole hydrocarbon production, such as fracturing operations.

[0028] Upon exiting acid scrubber system **86**, the pressure of stream **42** is increased by fan **100** and introduced into indirect heat exchanger **102**. Indirect heat exchanger **102** is of a variety commonly known to those skilled in the art of heat transfer. Preferably, ground water at approximately 55° F. is used to condense water vapor from stream **42**. The condensation of water vapor also assists in the removal of any remaining contaminates in the gas stream. Additionally, a condensate stream **104** comprising the water and any residual contaminants is returned to acid scrubber system **86** where it is combined with the brine.

[0029] Carbon dioxide stream **46** from the indirect heat exchanger **102** is introduced into carbon dioxide recovery unit **18**. Initially, stream **46** enters a refrigeration heat exchanger **108**. Stream **46** then enters carbon dioxide recovery system **110** where liquid carbon dioxide is separated from any excess oxygen or nitrogen remaining in stream **46**. Carbon dioxide recovery system **110** is of a design commonly known to those of ordinary skill in the art. As can be seen, liquid carbon dioxide stream **48** can be marketed as a saleable product. Finally, gas discharge stream **50** comprising excess oxygen and any nitrogen originally introduced through hydrocarbon feedstock streams **21** and **30** can be discharged to the atmosphere. Alternatively, the excess oxygen may be reused within the process as an oxidant or separated for bottling and sale as a product gas. Likewise, the excess nitrogen may be reused within the process as a gaseous fire blanket at the feedstock input or separated for bottling and sale as a product gas. When the process is operated under the conditions described herein, gas discharge stream **50** is eliminated or substantially reduced in comparison to prior art combustion processes.

[0030] By utilizing pure oxygen for gasification and oxidation as well as employing water injection and recirculation gas to moderate reaction temperatures, a preferred embodiment of the invention allows virtually all reaction products from the secondary combustion chamber **20** to be reused or marketed. These reaction products include carbon dioxide, water, and excess oxygen. In a preferred embodiment of the invention, provision is made to maintain the highest possible gasification/oxidation efficiency in order to reduce the level of trace organic compounds in the reaction products. Provision is also made to remove, with high efficiency, any acidic and particulate constituents produced by the combustion of less than ideal hydrocarbon feedstocks in the process, thereby allowing the recovery of reusable and marketable reaction products.

[0031] As shown in FIG. **1**A, the heat energy created and recovered from the system is used in an energy recovery boiler **14**, such as a heat recovery steam generator, to generate high and/or low pressure steam. The high and/or low pressure steam from the energy recovery boiler **14** (FIG. **1**A) is preferably used in a steam turbine **130** to generate electrical power via generator **132**. As shown in FIG. **1**A, steam leaving the steam turbine **130** is condensed in condenser **136** and returned as boiler feedwater **134** to the energy recovery boiler **14** via boiler feedwater preheater **138**. The generation of steam and power from heat energy is well known in the art and will not be further discussed herein. The overall power production process is much more energy efficient than conventional systerns/processes, because the natural resources consumed in the system/process are minimized, the products produced therefrom are marketable, and virtually no atmospheric or water emissions from the process are released to the environment.

[0032] In an alternative preferred embodiment of the invention as shown in FIG. **2**, the product water from the process is collected from polishing unit **16**. The product water consists of the recovered water quench and/or the water product formed from the complete combustion of various hydrocarbon feedstocks in the primary **10** and secondary **20** chambers (FIG. **1**A). Preferably, as shown in FIG. **2**, the product water is sent as steam to an electrolyzer or electrolysis unit **150** wherein the water molecules are split into oxygen and hydrogen gases by an electric current. A heat exchanger **12** is optionally used to generate the steam feed to electrolyzer **150**. The steam feed preferably has a temperature of about 250° F. and a pressure of about 200 psi. Stream **38** (FIG. **1**A) may be used to heat the product water into steam via heat exchanger **12**. The electrolyzer or electrolysis unit **150** electrically charges the water vapor, which causes the water molecules to become unstable, break apart, and reform as oxygen and hydrogen gases. In essence, the product water is converted into oxygen and hydrogen gases through the input of additional energy in the form of low cost electricity. The electrolyzer preferably operates with a low voltage of +/−6 volts DC. The mixed oxygen and hydrogen gases and any entrained water vapor are then routed to a separation unit **152** for subsequent separation into their respective pure component gases.

[0033] A membrane separation unit **152** is preferred for use in separating the oxygen and hydrogen gases and any entrained water vapor. Membrane separation technology is well known in the art and will only be briefly described herein. Membrane separation technology is based on the differing sizes of gas molecules to be separated. The synthetic membrane of the membrane unit **152** is arranged and designed so that its membrane pores are sized large enough to allow the desired gas molecules to pass through the membrane pores while preventing the passage of undesired gas molecules. Because membrane separation units are passive gas separation systems, the units produce relatively high purity gas separations but at much lower expense than other gas separation technologies. While membrane separation technology is preferred, other gas separation technologies, such as pressure swing adsorption, etc., may be equally employed to remove any entrained water vapor as well as separate the oxygen and hydrogen gases generated in the electrolysis unit **150**.

[0034] As shown in FIG. **2**, the membrane separation unit **152** preferably comprises at least three separate membranes **154**, **156**, **158**. The first membrane **154** of the membrane separation unit **152** is arranged and designed with membrane pores sized to allow the smaller hydrogen and oxygen gas molecules to pass therethrough while retaining, and thus separating, any entrained larger water vapor molecules. The recovered water vapor is collected and routed back to the process for subsequent reuse. The second membrane **156** of the membrane separation unit **152** is arranged and designed with membrane pores sized to allow hydrogen gas molecules to pass therethrough while retaining, and thus separating, the larger oxygen gas molecules.

[0035] The operating temperature (i.e., preferably above the dew point of water) and pressure (i.e., preferably above atmospheric) of the three membranes **154**, **156**, **158** are optimized to achieve the desired degree of separation. Likewise, a vacuum (not shown) may be drawn on the backside of the membranes **154**, **156**, **158** in order to enhance the recovery of desired molecules through the membranes **154**, **156**, **158**. While three separate membranes **154**, **156**, **158** are described herein, the invention is not limited to any particular number or arrangement of membranes. As is well known in the art, two or more membranes may be arranged in a series and/or parallel structure to produce the desired component separation and/or purification standards.

[0036] After an optional polishing and/or compression step **160**, the purified hydrogen gas **168** can bottled or otherwise sold as value added product. The recovered oxygen gas is further purified by sending the gas through third membrane **158** which prevents the passage of any additional water vapor that may be present as a result of water reformation. The recovered water vapor is collected and routed back to the process for subsequent reuse. After an optional polishing and/or compression step **162**, the purified oxygen gas **166** can be bottled or otherwise sold as value added product or routed back to the process as a feedstock in the primary **10** and/or secondary **20** combustion chambers.

[0037] In another embodiment of the system as shown in FIG. **2**, purified hydrogen gas **168** and the purified oxygen gas **166** generated a combustion process such as disclosed herein or in the inventor's prior art patents (also known as ZEROS plants) can be utilized in hydrogen fuel cells **170** to generate electric power **174**. Reacting a fuel, such as hydrogen, and oxygen electrochemically involves delivering fuel to a set of negative electrodes called anodes and delivering oxygen to a set of positive electrodes called cathodes inside the fuel cell **170**. As will be now be recognized by those of skill in the art, typical commercial carbonate fuel cells would be modified for this application. These prior art carbonate fuel cells are typically designed for methane (CH4) as the fuel and therefore generate carbon dioxide (CO2). As can now be seen, the inlet and outlets of the fuel cells **170** will be modified to input hydrogen stream **168** and output water stream **172**. Additionally, as discussed below, the anodes and cathodes will preferably be modified from the prior art.

[0038] The electrochemical reaction of hydrogen inside fuel cell **170** produces electrons. The electrochemical reaction of oxygen consumes electrons. Connecting the two produces a current of usable electric power **174**. Fuel cells **170** generally contain a layer between the electrodes called an electrolyte layer. The electrolyte has ions that move between the fuel and oxygen electrodes to keep the charge neutral between the electrodes as they produce and consume electrons. The electrolyte is preferably made from potassium and lithium carbonates and carbonate ions migrate between the fuel and oxygen.

[0039] Fuel cells **170** preferably are configured in stacks of individual cells connected in series. Fuel cell stacks preferably have up to 400 cells per stack and produce between 250 kW and 400 kW of electric power **174**. Fuel cells **170** preferably operate at a relatively high temperature (400 to 1,000 degrees F.). Operating at higher temperature allows the electrode reaction to produce power efficiently without expensive platinum type catalyst. It is also a high enough temperature that allows the hydrogen in the fuel stack to generate electricity. In addition to allowing hydrogen and oxygen in the fuel cell to react the high temperature of the process allows the fuel cell to reach a 60% efficiency and considering the combined heat and power (CHP) the produced energy efficiency can reach 90% or higher.

[0040] A fuel cell **170** of standard MW (megawatt) scale module contains four stacks nets around 1.4 MW of electric power **174** and can power sites like universities and hospitals as well as industrial sites and chemical plants. The modular design of fuel cell plants **170** let them scale up to a site's energy needs. When paired with a ZEROS plant the standard fuel cell size can deliver 59 MW's of electric power that is carbon free, is base load, and emits only a water stream **172**. This can be achieved in a ZEROS plant from the disposal of the municipal solid waste produced by the facility or could be fueled by plastic production waste or refinery waste from industry. This embodiment of a ZEROS plant combined with utilization of the hydrogen gas stream **168** and the oxygen steam **166** creates a waste to energy facility would solve several industry problems while producing base load energy with total carbon capture. This combustion free process emits only water, no pollutants, supporting the goal of no carbon emissions in generating power.

[0041] While the terms used herein are believed to be well-understood by one of ordinary skill in the art, definitions are set forth to facilitate explanation of certain of the presently-disclosed subject matter.

[0042] Following long-standing patent law convention, the terms “a”, “an”, and “the” refer to one or more when used in this application, including the claims. Thus, for example, reference to “a window” includes a plurality of such windows, and so forth.

[0043] Unless otherwise indicated, all numbers expressing quantities of elements, dimensions such as width and area, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about”. Accordingly, unless indicated to the contrary, the numerical parameters set forth in this specification and claims are approximations that can vary depending upon the desired properties sought to be obtained by the presently-disclosed subject matter.

[0044] As used herein, the term “about,” when referring to a value or to an amount of a dimension, area, percentage, etc., is meant to encompass variations of in some embodiments plus or minus 20%, in some embodiments plus or minus 10%, in some embodiments plus or minus 5%, in some embodiments plus or minus 1%, in some embodiments plus or minus 0.5%, and in some embodiments plus or minus 0.1% from the specified amount, as such variations are appropriate to perform the disclosed methods or employ the disclosed compositions.

[0045] The term “comprising”, which is synonymous with “including” “containing” or “characterized by” is inclusive or open-ended and does not exclude additional, unrecited elements or method steps. “Comprising” is a term of art used in claim language which means that the named elements are essential, but other elements can be added and still form a construct within the scope of the claim.

[0046] As used herein, the phrase “consisting of” excludes any element, step, or ingredient not specified in the claim. When the phrase “consists of appears in a clause of the body of a claim, rather than immediately following the preamble, it limits only the element set forth in that clause; other elements are not excluded from the claim as a whole.

[0047] As used herein, the phrase “consisting essentially of limits the scope of a claim to the specified materials or steps, plus those that do not materially affect the basic and novel characteristic(s) of the claimed subject matter. With respect to the terms “comprising”, “consisting of”, and “consisting essentially of”, where one of these three terms is used herein, the presently disclosed and claimed subject matter can include the use of either of the other two terms.

[0048] As used herein, the term “and/or” when used in the context of a listing of entities, refers to the entities being present singly or in combination. Thus, for example, the phrase “A, S, C, and/or

O” includes A, S, C, and O individually, but also includes any and all combinations and subcombinations of A, S, C, and O.

[0049] The Abstract of the disclosure is written solely for providing the United States Patent and Trademark Office and the public at large with a means by which to determine quickly from a cursory inspection the nature and gist of the technical disclosure, and it represents one preferred implementation of the invention and is not indicative of the nature of the invention as a whole.

[0050] It will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. Other embodiments of the invention will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. The foregoing disclosure and description are illustrative and explanatory thereof, and various changes in the details of the illustrated apparatus and construction and method of operation may be made without departing from the spirit in scope of the invention which is described by the following claims.

Claims

1. A process for treating waste materials to generate electric power comprising: introducing a waste stream, a first oxygen stream, a first fuel stream and a first water stream into a primary combustion chamber operated under vacuum conditions; reacting the waste, oxygen, fuel and water streams in the primary combustion chamber to produce a gaseous product stream comprising hydrogen, carbon dioxide, water vapor and other hydrocarbons and a solids stream; introducing the gaseous product stream into a cyclone separator to separate gaseous components of the gaseous product stream from any retained particulate components of the gaseous product stream, thereby creating a clean gaseous product stream; introducing the clean gaseous product stream into a secondary combustion chamber along with a second fuel stream, a second oxygen stream, and a second water stream, and thermally oxidizing the introduced streams to produce a reaction product stream comprising carbon dioxide and water; recycling a portion of the reaction product stream to the secondary combustion chamber for additional oxidation of any uncombusted reactants; introducing the remaining portion of the reaction product stream into an acid wash scrubber to remove residual fluorine, thereby creating a scrubbed reaction product stream; and separating the scrubbed reaction product stream into a brine stream comprising water and a scrubbed gaseous stream comprising carbon dioxide; removing a portion of the water from the brine stream creating a regenerated water stream; introducing the regenerated water stream into an electrolysis unit thereby electrolyzing the regenerated water stream to produce a hydrogen and oxygen containing stream; separating the hydrogen from the oxygen in the hydrogen and oxygen containing stream thereby creating a purified hydrogen stream and a purified oxygen stream; and introducing the purified hydrogen stream and the purified oxygen stream into a hydrogen fuel cell to generate electric power.

2. The process of claim 1 wherein said hydrogen fuel cell comprises a plurality of fuel cell stacks, each hydrogen fuel cell stack comprising a plurality of individual cells connected in series, each individual cell comprising an anode, an electrolyte, a cathode, a hydrogen inlet, an oxygen inlet, and a water outlet.

3. The process of claim 2 wherein said hydrogen fuel cell operates at between about 400 and about 1000° F.

4. The process of claim 2 wherein said hydrogen fuel cell operates at least 60% efficiency.

5. The process of claim 2 wherein the hydrogen fuel cell comprises four fuel cell stacks.

6. The process of claim 2 wherein the hydrogen fuel cell produces between 250 and 400 kW of electric power.

7. The process of claim 1 wherein the separating the hydrogen and oxygen step comprise introducing the hydrogen and oxygen containing stream into a membrane separation unit comprising a first membrane for separating water from the hydrogen and oxygen in the hydrogen

and oxygen containing stream, and a second membrane for separating hydrogen from oxygen.
8. The process of claim 2 wherein the electrolyte comprises potassium carbonate.
9. The process of claim 2 wherein the electrolyte comprises lithium carbonate.
10. The process of claim 2 wherein the hydrogen fuel cell comprises four hydrogen fuel cell stacks and produces about 1.4 megawatts of electric power.
11. The process of claim 1 wherein the waste stream comprises municipal solid waste.
12. The process of claim 1 wherein the waste stream is a liquid stream comprising sludge from refinery or chemical plants.
13. A process for treating waste materials to generate electric power comprising: introducing a waste stream, a first oxygen stream, a first fuel stream and a first water stream into a primary combustion chamber operated under vacuum conditions; reacting the waste, oxygen, fuel and water streams in the primary combustion chamber to produce a gaseous product stream comprising hydrogen, carbon dioxide, water vapor and other hydrocarbons and a solids stream; introducing the gaseous product stream into a cyclone separator to separate gaseous components of the gaseous product stream from any retained particulate components of the gaseous product stream, thereby creating a clean gaseous product stream; introducing the clean gaseous product stream into a secondary combustion chamber along with a second fuel stream, a second oxygen stream, and a second water stream, and thermally oxidizing the introduced streams to produce a reaction product stream comprising carbon dioxide and water; recycling a portion of the reaction product stream to the secondary combustion chamber for additional oxidation of any uncombusted reactants; introducing the remaining portion of the reaction product stream into an acid wash scrubber to remove residual fluorine, thereby creating a scrubbed reaction product stream; and separating the scrubbed reaction product stream into a brine stream comprising water and a scrubbed gaseous stream comprising carbon dioxide; removing a portion of the water from the brine stream creating a regenerated water stream; introducing the regenerated water stream into an electrolysis unit thereby electrolyzing the regenerated water stream to produce a hydrogen and oxygen containing stream; separating the hydrogen from the oxygen in the hydrogen and oxygen containing stream thereby creating a purified hydrogen stream and a purified oxygen stream; and introducing the purified hydrogen stream and the purified oxygen stream into a hydrogen fuel cell to generate electric power, wherein the hydrogen fuel cell comprises a plurality of fuel cell stacks, each fuel cell stack comprising a plurality of individual cells connected in series, each individual cell comprising an anode, an electrolyte, a cathode, a hydrogen inlet, an oxygen inlet, and a water outlet, and wherein the hydrogen fuel cell operates at between about 400 and 1000° F.
14. The process of claim 2 wherein said hydrogen fuel cell operates at least 60% efficiency.
15. The process of claim 2 wherein the hydrogen fuel cell comprises four fuel cell stacks.
16. The process of claim 2 wherein the hydrogen fuel cell produces between 250 and 400 kW of electric power.
17. The process of claim 1 wherein the separating the hydrogen and oxygen step comprise introducing the hydrogen and oxygen containing stream into a membrane separation unit comprising a first membrane for separating water from the hydrogen and oxygen in the hydrogen and oxygen containing stream, and a second membrane for separating hydrogen from oxygen.
18. The process of claim 2 wherein the electrolyte comprises potassium carbonate.
19. The process of claim 2 wherein the electrolyte comprises lithium carbonate.
20. The process of claim 2 wherein the hydrogen fuel cell comprises four hydrogen fuel cell stacks and produces about 1.4 megawatts of electric power.